Physics-Aware Feature Calibration Network for Remote Sensing Image Segmentation

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***Abstract*—In remote sensing image segmentation, regions with sharp intensity variations caused by complex illumination and ground objects significantly constrain segmentation accuracy. Most existing methods attempt to enhance the model's perception of these regions by introducing frequency features or relying on spatial convolutions. However, these methods tend to capture signal intensity, exhibiting limitations in distinguishing between sharp shadows and object edges in remote sensing images, which share similar intensities but stem from distinct physical origins. Through theoretical analysis, we observe that real edges typically exhibit an anisotropic gradient distribution, whereas shadow edges tend to show pseudoisotropic local gradient statistics under texture perturbations. To this end, we propose physics-aware feature calibration network (PAFCNet). The physics calibration module (PCM) introduces a GLRT-inspired differentiable decision mechanism to transform the anisotropy analysis of local gradient second moment into a learnable statistical decision process, thereby adaptively enhancing the generalizability of theoretical analysis in real-world scenes. In parallel, the frequency analysis module (FAM) constructs multidirectional object representations via affine transformations (ATs) and parameterized kernel functions (PKFs), while filtering out the interference of sharp shadows through the PCM. Finally, the dual domain fusion module (DFM) performs cross-sequence interaction between the physics-rectified frequency features and spatial features, thereby alleviating the semantic ambiguity associated with single-domain features. Experimental results demonstrate that PAFCNet outperforms state-of-the-art methods. The source code is available at https://github.com/wangjunyi-1/PAFCN**

***Index Terms*—Frequency-related learning, gradient statistics, remote sensing, state space model.**

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I. Introduction

With the development of aerospace sensors, remote sensing images have enabled fine-grained observation of the Earth's surface. Semantic segmentation, as a core technology for image interpretation, transforms raw pixels into structured geographic objects. The accuracy and reliability of its practical applications, such as land-cover mapping [1], [2], change detection [3], [4], and environmental protection [5], [6], [7], [8], are commonly and severely constrained by regions in the image with significant intensity variations.

These regions primarily originate from two distinct physical processes, specifically sharp shadow edges cast by terrain or elevated structures and abrupt changes in reflectance caused by edges between different ground objects. Despite their distinct physical origins, both phenomena manifest as sharp intensity variations in the image (as shown in Fig. 1). Consequently, models often misidentify shadow edges as object edges, leading to the semantic misclassification of shadow regions.

Current methods primarily enhance the discriminative ability of models for regions with sharp intensity variations by designing boundary-aware modules in the spatial domain or introducing frequency-domain analysis mechanisms. Huang et al. [9] improves segmentation through edge optimization, Li et al. [10] introduces a boundary-enhanced dual-stream architecture to refine object edges, and Zhou et al. [11] utilizes the wavelet transform to decompose images into low-frequency and high-frequency components and fuses these features to enhance boundary perception. Although these methods improve the perception of regions with significant intensity variations, they still fail to achieve satisfactory results in distinguishing regions with similar intensities but different physical origins.

Revisiting the challenge of sharp intensity variations, we identify that the core issue lies in the fact that existing methods tend to focus on capturing signal magnitude [12], [13], [14], [15]. Given that relying solely on signal magnitude is insufficient to effectively distinguish these confounded regions, a natural question arises as to whether we can capture the physical nature of the signal variation rather than merely staying at its superficial intensity. From the perspective of physical origins, object edges are formed by the junction of different objects, manifesting as signal transitions with a clear direction within a local neighborhood [16]. In contrast, although sharp shadows also exhibit sharp intensity transitions

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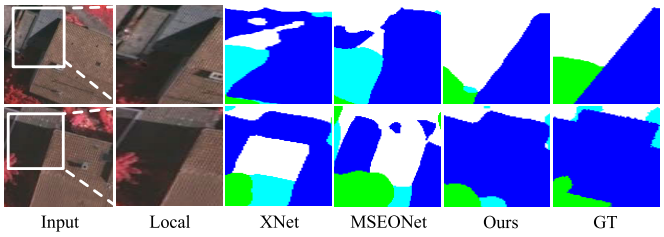


Fig. 1. Comparison of shadow robustness between frequency-domain (XNet), spatial-domain (MSEONet), and the proposed enhancement methods. The proposed method effectively mitigates misclassification caused by sharp intensity variations in shadow regions.

at their edges, they are compensated by ambient scattered light. Consequently, their contrast is insufficient to completely suppress the fluctuations of surface materials [17], [18]. Influenced by background texture perturbations, the directional consistency of shadow-edge responses may be weakened, making their local gradient statistics tend to exhibit a pseudoisotropic pattern. The confusion between sharp shadows and object edges is essentially an asymmetric mapping between change intensity and change patterns. Therefore, the key to resolving this problem lies in seeking high-order features capable of characterizing change patterns. We thereby extend our perspective from signal intensity to the gradient vector field and analyze its statistical morphology. This analysis relies on the spectral analysis of the local gradient second moment (Section III-B), aiming to transform implicit gradient distribution patterns into quantifiable anisotropy measures.

To this end, we propose physics-aware feature calibration network (PAFCNet), which consists of a feature encoder, physics calibration module (PCM), frequency analysis module (FAM), and domain fusion module (DFM). Unlike existing methods [9], [10], [19] that mainly enhance the perception of regions with significant intensity variations, we model such regions from their physical origins. Specifically, our analysis in Section III-B shows that sharp shadow edges are more likely to exhibit spectral expansion under texture perturbations, whereas real object edges retain stronger anisotropy. Motivated by this distinction, PCM employs a GLRT-inspired differentiable decision mechanism on local gradient spectral information to generate discriminative statistical weights. Considering the arbitrary orientations and diverse geometric structures of ground objects in overhead remote sensing images, FAM uses affine transformations (ATs) and parameterized kernel functions (PKFs) to adaptively capture frequency patterns of ground objects, while suppressing sharp-shadow interference with the aid of PCM. Finally, DFM performs cross-sequence interaction between frequency and spatial features to alleviate semantic ambiguity. In summary, our main contributions are as follows.

1) We systematically analyze the segmentation inaccuracies induced by sharp intensity variations and propose PAFCNet from the perspective of the physical origins of such regions to effectively mitigate this issue.
2) We elucidate through theoretical analysis that local gradient spectral features can quantify the anisotropic differences between sharp shadows and object edges. On this basis, we design the PCM to transform spectral features into a learnable statistical decision process, enhancing the generalizability of theoretical analysis in real-world scenes.
3) We design the FAM, which combines affine-aligned sampling with parameterized sinusoidal kernels to perform geometry-adaptive frequency-related modeling of object structures, thereby improving the representation of directional textures and local geometric details.

II. Related Work

A. Boundary-Aware and Detail Enhancement

To improve the discriminative capability for regions with sharp intensity variations in remote sensing image segmentation, existing studies mainly proceed along two lines. One line strengthens the structural representation of contour regions through boundary modeling or boundary enhancement mechanisms. Specifically, MSEONet [9] adaptively refines target contours through an edge optimization strategy to enhance the discriminability of boundary regions; BEDSN [10] adopts a boundary-enhanced dual-stream architecture to compensate for the loss of boundary information in high-resolution remote sensing images beyond the semantic branch. The other line mainly improves local structural representation and semantic expression through detail recovery and context modeling, thereby indirectly alleviating the misclassification caused by regions with sharp intensity variations [20], [21], [22], [23]. For example, RSSFormer [19] strengthens foreground saliency through background noise suppression and a detail-aware attention mechanism, thereby improving detail representation and boundary quality in complex scenes. Compared with explicit boundary enhancement methods, this category of methods does not directly model edges separately, but instead indirectly improves the model's ability to distinguish target contours and fine-grained structures through the collaborative optimization of local detail features and contextual features. In addition to boundary and detail enhancement, recent foundation-model-based studies also show the importance of task-oriented adaptation and remote sensing-specific priors in dense prediction. Meta-CD [24] effectively adapts FastSAM to limited-data change detection, while RSKT-Seg [25] enhances open-vocabulary remote sensing segmentation with domain-specific knowledge transfer.

Overall, although existing methods [26], [27], [28], [29] have achieved significant progress in representation capability, most of them still mainly rely on data-driven feature fitting and appearance modeling, and lack explicit characterization of the physical origins behind regions with sharp intensity variations.

B. Frequency-Related Learning

As a useful complement to spatial-domain representations, frequency-related analysis has gradually been introduced into deep models. Related methods typically employ frequency analysis techniques such as the wavelet transform [30], discrete cosine transform (DCT) [31], and Gabor [32] filtering to

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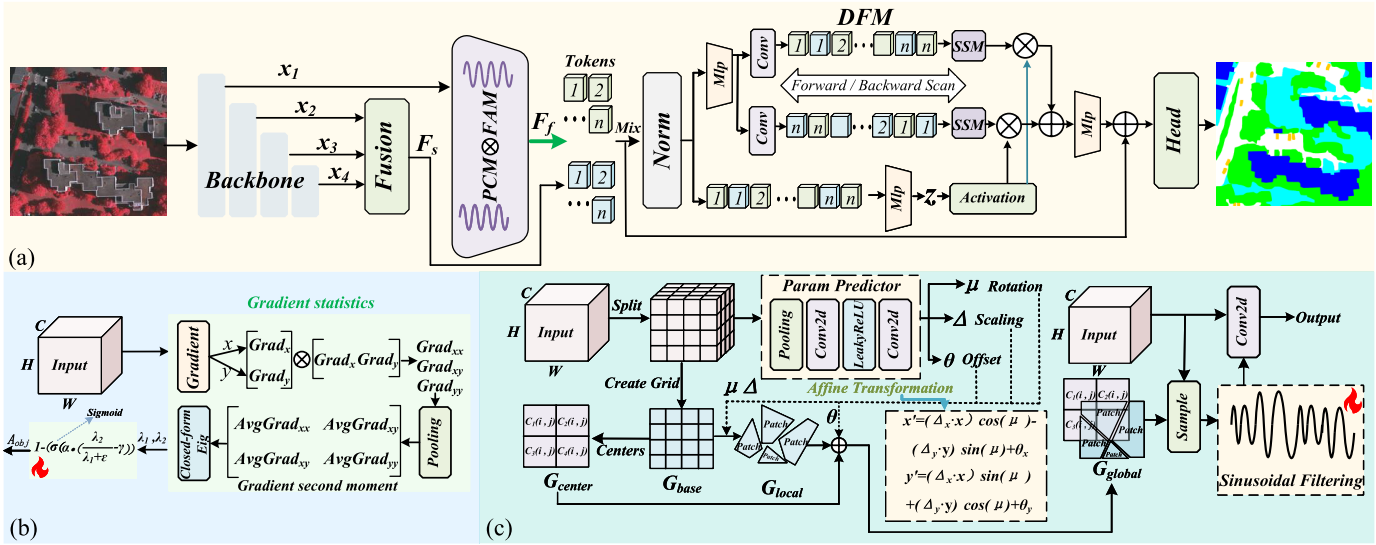


Fig. 2. Overall architecture of PAFCNet. The feature x_1 from the backbone network is input into the PCM and FAM to obtain the enhanced feature F_f via element-wise multiplication. Subsequently, F_f undergoes interactive fusion with the deep feature sequence F_s in the DFM, and the prediction map is finally output through the segmentation head. (a) PAFCNet. (b) PCM. (c) FAM.

transform images or features, or to model their responses, so as to mine pattern information that is difficult to explicitly express in the spatial domain [31], [33], [34], [35]. Early studies mainly focused on introducing frequency decomposition mechanisms into network design to enhance the model's perception of different frequency components. For example, XNet [11] uses the wavelet transform to decompose the input into low-frequency and high-frequency components and models them with a dual-branch architecture. As frequency-related representation learning has been more deeply explored in remote sensing image segmentation, the research focus has gradually extended from pure frequency decomposition to the collaborative modeling of frequency-related and spatial-domain information. This is because relying solely on spatial-domain features often makes it difficult to simultaneously capture the overall structure of a scene and its local details, whereas frequency-related representations can provide additional structural and textural cues. Based on this, AFENet [36] designs an adaptive interaction mechanism between frequency-domain and spatial-domain features. GFDNet [37] demonstrates the effectiveness of learnable Fourier-domain filtering in capturing global contextual information for high-resolution remote sensing segmentation, while F2Net [38] further demonstrates the advantage of high- and low-frequency decomposition in preserving fine structural details for remote sensing segmentation.

Although frequency-related representation learning provides complementary structural and textural cues, most existing studies mainly exploit frequency components or frequency-inspired responses for feature enhancement. However, they are not explicitly designed to determine whether a high-frequency or edge-like response originates from a physical object boundary or from an illumination-induced shadow edge. As a result, frequency-based enhancement alone may amplify shadow-related pseudoedges together with real object structures.

III. Methodology

A. PAFCNet Architecture

The overall architecture of PAFCNet is illustrated in Fig. 2. We employ a pretrained ConvNeXt [39] as the backbone to extract multiscale features from the input image. The feature x_1 is fed into both the PCM and the FAM, while the deeper features x_2, x_3, and x_4 are aggregated into F_s. To obtain discriminative evidence capable of distinguishing sharp shadows from object edges, the PCM performs a GLRT-inspired differentiable decision mechanism on local gradient statistics, generating statistical weights that adaptively quantify structural confidence. In parallel, the FAM branch utilizes ATs and PKFs to adapt to the multidirectional geometric patterns of input features, extracting dynamic frequency representations rich in texture details to enhance the model's perception of regions with significant intensity variations. Subsequently, the weight map generated by the PCM modulates the frequency representations from the FAM via element-wise multiplication, filtering out the interference of sharp shadows. Finally, to address the semantic ambiguity of single-domain features, the DFM performs cross-sequence interaction between the physics-calibrated frequency features F_f and the spatial features F_s, outputting the final segmentation map.

B. Anisotropy Analysis of Shadow and Object Edges

Significant intensity variations in remote sensing images stem from both sharp shadow edges and object edges. While existing methods often focus solely on modeling object edges, the limited penumbra regions of shadows cast by man-made structures in remote sensing images result in sharp intensity discontinuities. These discontinuities exhibit high-frequency characteristics in signal magnitude that are almost consistent with object edges, making it difficult for traditional intensity-based methods to effectively distinguish between

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them. Subsequently, we analyze the gradient statistical properties under these two scenarios and provide a statistical basis for the discriminative design of PCM through the spectral analysis of the local gradient second moment.

Let the image gradient field ∇I within a local window Ω be modeled as a superposition of two components, specifically a deterministic structural component ∇I_s originating from object or shadow edges and a stochastic texture component ∇I_t derived from surface material properties [40]

$$\nabla I(\mathbf{x}) = \nabla I_s(\mathbf{x}) + \nabla I_t(\mathbf{x}), \quad \mathbf{x} \in \Omega. \tag{1}$$

Divide the window Ω into N micro-subregions $\{\Omega_k\}_{k=1}^N$. The local gradient second moment of the kth subregion is defined as

$$\mathbf{S}_k = \sum_{\mathbf{x}\in\Omega_k} (\nabla I_s + \nabla I_t)(\nabla I_s + \nabla I_t)^\top. \tag{2}$$

Here, the local gradient second moment is obtained by accumulating local gradient outer products without mean centering. Based on the statistical orthogonality assumption [41], the structural gradient ∇I_s and the texture gradient ∇I_t are uncorrelated within the local window ($\mathbb{E}[\nabla I_s \nabla I_t^\top] \approx \mathbf{0}$). Consequently, the cross terms in the expansion can be neglected, yielding the simplified model

$$\mathbf{S}_k \approx \mathbf{S}_{s,k} + \mathbf{S}_{t,k} \tag{3}$$

where $\mathbf{S}_{s,k}$ is a rank-1 structural second moment with energy σ_s^2 and principal direction $\mathbf{u}_s$, and $\mathbf{S}_{t,k}$ is the texture second moment.

To better approximate real-world ground features, the texture second moment $\mathbf{S}_{t,k}$ is modeled as an anisotropic random process with eigenvalues $\sigma_{t,1}^2 \geq \sigma_{t,2}^2 \geq 0$ and principal direction $\mathbf{u}_{t,k}$. We define the texture anisotropy strength as $\Delta\sigma_t^2 = \sigma_{t,1}^2 - \sigma_{t,2}^2$ and the angle between texture and structure directions as $\delta_k = \angle(\mathbf{u}_{t,k}, \mathbf{u}_s)$. In the local coordinate system of the structural principal direction $\mathbf{u}_s$, where $\mathbf{u}_s$ aligns with the x-axis, the structural second moment assumes a diagonal form

$$S_{s,k} = \begin{pmatrix} \sigma_s^2 & 0 \\ 0 & 0 \end{pmatrix}. \tag{4}$$

The texture second moment can be represented as a rotated diagonal matrix

$$S_{t,k} = R(\delta_k) \begin{pmatrix} \sigma_{t,1}^2 & 0 \\ 0 & \sigma_{t,2}^2 \end{pmatrix} R(\delta_k)^\top \tag{5}$$

$$R(\delta_k) = \begin{pmatrix} \cos\delta_k & -\sin\delta_k \\ \sin\delta_k & \cos\delta_k \end{pmatrix}. \tag{6}$$

After expansion, the off-diagonal element of the total second moment $S_k = S_{s,k} + S_{t,k}$ is

$$(S_k)_{12} = \frac{1}{2}\Delta\sigma_t^2 \sin(2\delta_k). \tag{7}$$

According to matrix perturbation theory [42], the presence of texture leads to a rotation angle ϕ_k of the principal eigenvector. This rotation angle satisfies

$$\tan(2\phi_k) = \frac{2(S_k)_{12}}{(S_k)_{11} - (S_k)_{22}} = \frac{\Delta\sigma_t^2 \sin(2\delta_k)}{\sigma_s^2 + \Delta\sigma_t^2 \cos(2\delta_k)} \tag{8}$$

where the numerator $\Delta\sigma_t^2 \sin(2\delta_k)$ represents the perturbing force of the texture, while the denominator $\sigma_s^2 + \Delta\sigma_t^2 \cos(2\delta_k)$ represents the inertia of the structure.

As can be seen from (8), the deflection angle of the principal direction, ϕ_k, depends not only on the strength of the texture perturbation term $\Delta\sigma_t^2 \sin(2\delta_k)$, but also on the suppressive effect of the structural energy term $\sigma_s^2 + \Delta\sigma_t^2 \cos(2\delta_k)$ on such perturbation. For real object edges, since they mainly correspond to discontinuities in physical properties such as reflectance rather than pure illumination changes [43], the local gradients are more likely to be dominated by structural transitions. Brox et al. [44] and Field et al. [45] show that such edges are more likely to form a stable dominant direction within a local neighborhood and exhibit stronger directional consistency. In regions with significant intensity variations induced by real edges, we describe this case as one in which the structural gradient energy σ_s^2 is more likely to dominate over the texture term, or the principal texture direction tends to be more statistically aligned with the principal structural direction, making δ_k less likely to produce a significant perturbation term. For shadow edges, the corresponding structural term mainly reflects brightness transitions caused by illumination occlusion, rather than discontinuities in physical properties as in real object edges. Sato et al. [46], DiMattina et al. [47], and Tappen et al. [48] show that shadow regions do not completely lose the observed responses caused by surface reflectance and texture. Therefore, we describe shadow edges as a case in which the structural gradient energy σ_s^2 cannot stably suppress texture perturbations across different subregions, and the principal structural direction is less likely to maintain stable consistency with the local principal texture direction, causing the relative orientation angle δ_k to exhibit a more dispersed distribution across different subregions Ω_k.

The macroscopic gradient statistics within the local window can be obtained through pooling operations. Since average pooling only introduces a constant normalization factor under a fixed window size, it preserves the eigenvalue-ratio statistic used in PCM. Therefore, this aggregation can be expressed as the sum of the subregion second moments

$$\mathbf{S}_\Omega = \sum_{k=1}^N \mathbf{S}_k. \tag{9}$$

Compared with λ_1, which mainly reflects the principal-direction response, λ_2 reflects the accumulated gradient energy perpendicular to the structural principal direction $\mathbf{u}_s$, and more directly characterizes the spectral expansion induced by texture perturbations and directional inconsistency. Through orthogonal projection

$$\lambda_2(\mathbf{S}_\Omega) \approx \sum_{k=1}^N (\mathbf{S}_{t,k})_{22} = \sum_{k=1}^N \left(\sigma_{t,1}^2 \sin^2\delta_k + \sigma_{t,2}^2 \cos^2\delta_k\right). \tag{10}$$

Using the definition $\Delta\sigma_t^2 = \sigma_{t,1}^2 - \sigma_{t,2}^2$ and the trigonometric identity $\sin^2\delta_k + \cos^2\delta_k = 1$, we derive the spectral expansion formula

$$\lambda_2(\mathbf{S}_\Omega) \approx \sum_{k=1}^N \left(\left(\sigma_{t,2}^2 + \Delta\sigma_t^2\right)\sin^2\delta_k + \sigma_{t,2}^2 \cos^2\delta_k\right)$$

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$$= \sum_{k=1}^{N} \left(\sigma_{t,2}^2 + \Delta\sigma_t^2 \sin^2 \delta_k\right)$$

$$= N\sigma_{t,2}^2 + \Delta\sigma_t^2 \sum_{k=1}^{N} \sin^2 \delta_k \tag{11}$$

where $N\sigma_{t,2}^2$ represents the noise floor, and $\Delta\sigma_t^2 \sum_{k=1}^{N} \sin^2 \delta_k$ denotes the additional orthogonal energy accumulated from texture-structure misalignment. For object edges, the spectral expansion term remains relatively small due to either energy dominance, where $\Delta\sigma_t^2$ is relatively negligible compared with the structural gradient energy, or directional consistency, where $\sin^2 \delta_k$ tends to be small in most subregions. As a result, λ_2 depends primarily on the noise floor. In contrast, for shadow edges, the weaker structural dominance makes it difficult to stably suppress texture perturbations across different subregions, and the relative orientation δ_k is less likely to remain concentrated around zero. Consequently, $\sin^2 \delta_k$ tends to take non-negligible values in more subregions, which generates a stronger spectral expansion effect after aggregation and elevates λ_2.

C. Physics Calibration Module

Based on the above analysis, the distinction between sharp shadow edges and real object edges can be attributed to the difference in the spectral features of the local gradient second moment. Inspired by the ratio-based decision principle of the generalized likelihood ratio test, we model their distinction as a binary hypothesis testing problem and define the ratio of λ_2 to λ_1 as the anisotropy statistic to reduce the influence of overall edge strength on discrimination

$$T(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}) + \epsilon} \tag{12}$$

where ϵ is a numerical stability constant. T quantifies the degree of spectral expansion. Considering that theoretical derivations rely on ideal statistical assumptions, which may deviate from complex real-world distributions, we transform this theoretical analysis into a learnable decision process. Specifically, the physical structure confidence map A_{obj} is calculated as follows:

$$\mathcal{A}_{obj}(\mathbf{x}) = 1 - \sigma(\alpha \cdot (T(\mathbf{x}) - \tau)) \tag{13}$$

where $\sigma(\cdot)$ denotes the Sigmoid function. τ represents the learned anisotropy threshold characterizing the tolerance limit for incoherent texture perturbations, while α serves as the decision sharpness to regulate the discrimination confidence regarding physical criteria by controlling the steepness of the activation function. In implementation, the eigenvalue statistics are computed in closed form from the local 2×2 gradient second moment, so that gradients can be backpropagated through standard differentiable operators. For numerical stability, a small constant ϵ is introduced in (12) to avoid division by zero, and is additionally used to regularize the square-root term in the closed-form eigenvalue computation. The computed eigenvalues are further constrained with a small lower bound to avoid invalid values caused by finite-precision errors. Moreover, α is parameterized by Softplus$(\cdot)$ to keep the decision sharpness positive during training.

D. Frequency Analysis Module

Due to the inherent overhead imaging characteristics of remote sensing images, ground objects appear in arbitrary orientations, whereas existing frequency-related analysis methods usually adopt fixed analysis patterns, making it difficult to stably adapt to complex variations in object orientation and geometry. Different from the above principled frequency analysis schemes, we employ affine-aligned sampling to geometrically normalize local structures, so that they are mapped into a more regular analysis coordinate system, and introduce a parameterized sinusoidal kernel on the aligned features to perform frequency-related modeling of the textural organization and geometric details in arbitrarily oriented objects.

Specifically, we first construct a sampling grid that matches the object's shape using an AT. To this end, we partition the input x_1 into a set of local feature patches P_i. For each feature patch P_i, we utilize a parameter predictor F_{pred} to generate a set of deformation parameters

$$(s_i, \theta_i, \Delta p_i) = F_{pred}(P_i) \tag{14}$$

where s_i represents the nonuniform scaling factors, θ_i is the rotation angle, and Δp_i is the local offset. Next, based on the predicted parameters, we define an AT matrix

$$A_i = \begin{pmatrix} s_{ix}\cos\theta_i & -s_{iy}\sin\theta_i \\ s_{ix}\sin\theta_i & s_{iy}\cos\theta_i \end{pmatrix} \tag{15}$$

where A_i is the AT matrix. Applying A_i to the standardized local coordinate grid P_{local} yields the transformed local sampling grid G_i^{local}

$$G_i^{local} = A_i P_{local}. \tag{16}$$

The transformed local sampling grid is then placed at its global position to form the final sampling grid, which is used to sample from the feature map x_1

$$q = p_{center} + \Delta p_i + G_i^{local} \tag{17}$$

$$F_{align} = GridSample(q, x_1) \tag{18}$$

where q represents the final sampling point coordinates, p_{center} is the global center coordinate corresponding to the feature patch, and $GridSample(\cdot)$ represents the dynamic sampling operation. On the aligned features, we further introduce a parameterized sinusoidal kernel to model the textural organization and geometric details of the objects

$$x' = x\cos(\theta_c) + y\sin(\theta_c) \tag{19}$$

$$K_c(x, y) = \cos\left(\frac{2\pi x'}{\lambda_c} + \psi_c\right) \tag{20}$$

where for c channels, we define an independent set of learnable parameters: the orientation angle θ_c, wavelength λ_c, and phase ψ_c. These control the filter's orientation, response frequency, and symmetry, respectively. During training, these parameters are constrained within stable ranges through parameterized mappings. Finally, the filtering operation is completed using a group convolution

$$F_a = F_{align} * K_c(x, y) \tag{21}$$

where F_a is the final output of the FAM, and $*$ represents the group convolution operation. The physical structure

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confidence map A_{obj} derived from the PCM modulates the frequency feature map F_a through element-wise multiplication to yield F_f.

E. Dual DFM

In remote sensing images, regions with significant intensity variations are often continuously distributed in urban scenes, while they tend to appear more locally in rural and similar scenes. Different regions may rely on frequency details and spatial semantics to different degrees. Relying on a single feature source is insufficient to adapt to such regional differences and may lead to semantic ambiguity. The selective update mechanism of Mamba [49], [50], [51] can dynamically regulate the roles of different features during interaction according to the input content, which helps coordinate frequency details and spatial semantics. Unlike existing methods [29], [52] that mainly use Mamba to integrate global contextual information or perform cross-stage feature interaction, we serialize heterogeneous features at the same spatial location to exploit their complementarity. As the spatial scan proceeds, these heterogeneous features can interact within a larger context, thereby gradually alleviating the semantic ambiguity caused by a single feature source.

Specifically, for the input features F_f and F_s, we first cross-mix them at each spatial location into a sequence S_{mix}. Next, the mixed sequence S_{mix} is fed into two parallel branches. One branch utilizes an SSM to perform bi-directional modeling on it to capture deep dependencies, while the other branch feeds S_{mix} into an MLP and an activation function to generate weights Z for fusion. Finally, we use these weights Z to perform a weighted fusion of the features processed by the SSM branch, and then obtain the final output through another MLP and a residual connection. The process is formulated as follows:

$$S_{mix} = \left\{F_f^i, F_s^i\right\}_{i=1}^{n} = \left[F_f^1, F_s^1, F_f^2, F_s^2, \ldots,\right] \tag{22}$$

$$z_1, z_2 = SiLU\left(\mathcal{T}\left(S_{mix}\right)\right) \tag{23}$$

$$S'_{mix} = S_{mix} + \mathcal{T}\left(z_1 \cdot S_f + z_2 \cdot S_b\right) \tag{24}$$

where $\mathcal{T}(\cdot)$ denotes the MLP, $SiLU(\cdot)$ is the SiLU activation function, and S_f and S_b are the sequences generated from the bi-directional modeling.

F. Loss Function

For the PAFCNet proposed by us, the entire network is trained using a loss function $\mathcal{L}$, which consists of $\mathcal{L}_{ce}$ and $\mathcal{L}_{dice}$. Specifically

$$\mathcal{L}_{ce} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \tag{25}$$

$$\mathcal{L}_{dice} = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \frac{\hat{y}_k^{(n)} y_k^{(n)}}{\hat{y}_k^{(n)} + y_k^{(n)}} \tag{26}$$

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{dice} \tag{27}$$

where N denotes the number of training samples, and K denotes the number of classes. $y_k^{(n)}$ represents the one-hot encoding of the ground truth that sample $n \in [1, \ldots, N]$ belongs to class k, and $\hat{y}_k^{(n)}$ denotes the probability that sample n belongs to class k.

IV. Experiments

A. Experimental Settings

1) Datasets: For our evaluation, we used the Vaihingen, LoveDA, and Potsdam datasets, which are standard benchmarks for remote sensing semantic segmentation. These widely used datasets were selected to ensure the comparability and generalizability of our results, as they cover diverse land categories, such as buildings and road systems, under complex environmental conditions like varied seasons and lighting. Their prevalence in academia facilitates a direct comparison of our work with previous studies.

a) Vaihingen dataset: The dataset comprises 33 high-resolution TOP image tiles, each with an average dimension of 2494 $\times$ 2064 pixels, featuring real orthophoto (TOP), digital surface model (DSM), and normalized DSM (NDSM) data. It includes five foreground land cover categories: impervious surfaces, buildings, low vegetation, trees, and cars, along with a background class: clutter. During the experiment, we carefully chose training samples based on the specific training IDs (ID 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37) provided by the ISPRS competition, using the remaining 17 images for testing. To facilitate processing, these image tiles were cropped into smaller 1024 $\times$ 1024 pixel blocks.

b) Loveda dataset: The dataset is built from Google Earth imagery, spanning three Chinese cities: Nanjing, Changzhou, and Wuhan. It comprises 5987 high-resolution optical remote sensing images at 0.3 m ground resolution. Each 1024 $\times$ 1024 pixel image includes seven land cover categories: buildings, roads, water bodies, wasteland, forests, agricultural areas, and background. The dataset contains three spectral bands: red, green, and blue. Unlike the Vaihingen and Potsdam dataset processing, LoveDA divides training, validation, and test sets by geographical location to amplify regional diversity. Specifically, the dataset is split into 2522 training images, 1669 validation images, and 1796 official test images. It encompasses both urban and rural landscapes.

c) Potsdam dataset: The Potsdam dataset offers a more extensive and challenging evaluation scenario, containing 38 high-resolution tiles from Potsdam, Germany. Each 6000 $\times$ 6000 pixel tile features a fine-grained ground sampling distance (GSD) of 5 cm and provides data across four multispectral bands (red, green, blue, and near-infrared), accompanied by a corresponding DSM. According to the official ISPRS protocol, 24 tiles are used for training, while the remaining 14 tiles are used for testing. For processing, these large tiles are partitioned into 1024 $\times$ 1024 patches. Following common practice, the "clutter/background" class is excluded during quantitative evaluation.

d) Shadow-aware subset: To evaluate shadow-related errors more directly, we construct a shadow-aware subset using a deterministic rule based only on the input image and semantic ground-truth labels. Given an input image, we first compute the luminance map as

$$L(x) = 0.299R(x) + 0.587G(x) + 0.114B(x). \tag{28}$$

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The local background luminance $L_{bg}(x)$ is estimated by Gaussian smoothing and morphological closing, and the relative darkness response is computed as

$$D(x) = \frac{L_{bg}(x) - L(x)}{L_{bg}(x) + 10^{-6}}. \tag{29}$$

Pixels satisfying $L(x) \leq Q_{0.30}(L)$ and $D(x) \geq 0.15$ are selected as initial dark-region candidates. After removing ignore-label pixels and excluding water pixels for LoveDA, connected components are first filtered by area ratio

$$0.001 \leq \frac{|C_i|}{|\Omega|} \leq 0.35. \tag{30}$$

Since we focus on the confusion between shadow edges and real object boundaries, we further retain components adjacent to semantic boundaries

$$\frac{|C_i \cap \text{Dilate}(\partial Y, 15)|}{|C_i|} \geq 0.02. \tag{31}$$

The retained components form the candidate shadow mask S. Images satisfying $|S|/|\Omega| \geq 0.002$ are included in the shadow-aware subset.

2) Implementation Details: All experiments were conducted on a single NVIDIA GeForce RTX 2080Ti (11 GB) GPU, running on an Ubuntu 18.04 system with the PyTorch 1.11 framework. For model optimization, we employed the AdamW optimizer, setting the initial learning rate to 6×10^{-4} and decaying it throughout training via a cosine scheduling strategy. For the Vaihingen and Potsdam datasets, all images were preprocessed by partitioning them into 1024×1024 patches. During the training phase, a suite of data augmentation techniques was applied to enhance model generalization, including random horizontal/vertical flipping, random rotation, and random scaling with factors of (0.5, 0.75, 1.0, 1.25, and 1.5).

3) Evaluation Metrics: To quantitatively evaluate the proposed model, we adopt overall accuracy (OA), mean intersection over union (mIoU), and $F1$-score ($F1$). We also report Parameters (Params), floating-point operations (FLOPs), and inference time for efficiency evaluation. For the constructed shadow-aware subset, we further report shadow-region mIoU, nonshadow-region mIoU, shadow-boundary error, and false attachment degree (FAD).

Given the prediction P, semantic ground truth Y, and candidate shadow mask S, the shadow-region and nonshadow-region mIoU are computed as

$$\begin{aligned} \text{mIoU}_{sh} &= \text{mIoU}(P, Y \mid S) \\ \text{mIoU}_{non} &= \text{mIoU}\left(P, Y \mid \bar{S}\right). \end{aligned} \tag{32}$$

For a given region R, the regional mIoU is defined as

$$\text{mIoU}(P, Y \mid R) = \frac{1}{|\mathcal{C}_R|} \sum_{c \in \mathcal{C}_R} \frac{\left|(P = c) \cap (Y = c) \cap R\right|}{\left|((P = c) \cup (Y = c)) \cap R\right|} \tag{33}$$

where $\mathcal{C}_R$ denotes the set of valid classes whose union is nonempty in R.

For shadow-boundary evaluation, the boundary band and semantic error rate are computed as

$$B_{sh} = \text{Dilate}(\partial S, 15) \tag{34}$$

$$\text{Err}_{sh\text{-}bd} = 1 - \text{Acc}(P, Y \mid B_{sh}). \tag{35}$$

To measure whether candidate shadow regions are incorrectly predicted as adjacent semantic classes, we define the surrounding ring of the ith connected component S_i as

$$R_i = \text{Dilate}(S_i, 25) \setminus S_i. \tag{36}$$

Let C_i^{adj} be the set of valid adjacent semantic classes within R_i. FAD is defined as

$$\text{FAD} = \frac{\sum_i \left|S_i \cap \left(P \in C_i^{\text{adj}}\right) \cap (P \neq Y)\right|}{\sum_i |S_i|}. \tag{37}$$

B. Comparative Experiments

For our comparative analysis, we selected a diverse set of representative models related to sharp-intensity-variation modeling in remote sensing image segmentation. These include boundary- and edge-enhancement methods such as MSEONet [9], CPNet [54], and BEDSN [10], foreground enhancement methods such as RSSFormer [19], the recently prominent SSM-based architecture RSMamba [29], and frequency-domain enhancement methods including XNet [11] and AFENet [36]. Table I presents the detailed comparative results of PAFCNet against these methods on the Vaihingen, Potsdam, and LoveDA datasets.

The efficiency analysis in Table II shows that PAFCNet achieves favorable efficiency with moderate model complexity. Specifically, PAFCNet contains 28.42 M parameters, which is lower than those of XNet (326.18 M), RSSFormer (31.89 M), RSMamba (40.73 M), CPNet (36.59 M), and MSEONet (54.13 M). In terms of computational complexity, PAFCNet requires 131.94 G FLOPs, which is substantially lower than XNet (5343.39 G), BEDSN (980.46 G), MSEONet (516.79 G), RSSFormer (336.67 G), and CPNet (304.62 G). Its inference time reaches 89.15 ms, outperforming RSSFormer (117.44 ms), RSMamba (134.97 ms), BEDSN (179.45 ms), MSEONet (164.93 ms), AFENet (125.29 ms), and especially XNet (1269.87 ms). This favorable efficiency is partly attributed to our use of a closed-form eigenvalue computation. By computing eigenvalues from the trace and determinant rather than using iterative eigenvalue decomposition, we transform the operation into standard tensor computation and reduce the latency introduced by an iterative solver.

1) Results of the Vaihingen Dataset: In the experiments on the Vaihingen dataset (as shown in Table I), PAFCNet performs excellently on all key metrics, achieving a mean $F1$, mIoU, and OA of 91.62%, 84.70%, and 91.90%, respectively. This performance not only surpasses the Transformer-based UnetFormer but also significantly leads the SSM-based RSMamba. Compared to XNet, which also utilizes frequency-domain information for enhancement, PAFCNet is higher in $F1$, mIoU, and OA by 1.05%, 1.75%, and 0.69%, respectively.

This advantage in quantitative performance stems from the model's discriminative modeling of the physical origins leading to significant intensity variations. As shown in Fig. 3, all displayed samples visually exhibit abrupt illumination changes induced by building projections. Since sharp shadow edges and

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TABLE I

COMPARISON OF SEGMENTATION RESULTS ON THE VAIHINGEN, POTSDAM, AND LOVEDA DATASETS

Dataset	Metric(%) / Class		UnetFormer ISPRS22 [53]	XNet ICCV23 [11]	RSSFormer TIP23 [19]	RSMamba GRSL24 [29]	CPNet AEOG24 [54]	BEDSN GIS24 [10]	MSEONet TGRS25 [9]	AFENet TGRS25 [36]	PAFCNet (Ours)
Vaihingen	F1	Imp.Surf	92.70	92.89	92.84	93.10	92.81	93.02	89.91	93.33	**93.79**
		Building	95.30	95.48	95.69	95.61	95.36	95.45	93.09	95.56	**96.06**
		Lowveg	84.90	85.17	83.80	84.09	85.05	85.04	83.75	85.29	**86.26**
		Tree	90.60	**91.10**	89.96	90.26	90.76	90.45	90.09	90.87	91.07
		Car	88.50	88.22	89.30	89.42	89.35	86.70	79.33	**91.16**	90.90
	MeanF1		90.40	90.57	90.32	90.50	90.67	90.13	87.24	91.24	**91.62**
	mIoU		82.70	82.95	82.58	82.87	83.11	82.26	77.70	84.11	**84.70**
	OA		91.00	91.21	90.78	90.98	91.14	91.09	89.15	91.44	**91.90**
Potsdam	F1	Imp.Surf	92.37	91.98	92.52	92.56	90.94	91.83	88.34	93.47	**93.71**
		Building	96.21	96.80	96.51	96.63	95.15	95.65	91.40	96.49	**97.35**
		Lowveg	86.87	86.93	87.03	87.43	86.35	86.20	83.97	87.93	**88.77**
		Tree	88.24	88.76	88.75	88.51	88.45	87.68	86.46	88.73	**89.80**
		Car	95.31	95.81	95.35	95.72	94.78	95.55	91.56	96.59	**96.66**
	MeanF1		91.80	92.05	92.03	92.17	91.13	91.38	88.34	92.64	**93.26**
	mIoU		85.06	85.51	85.45	85.70	83.90	84.37	79.25	86.52	**87.57**
	OA		90.39	90.56	90.63	90.70	89.25	89.45	86.33	91.22	**91.88**
LoveDA	F1	Building	79.54	79.35	69.70	78.79	73.46	66.52	76.54	80.37	**82.21**
		Road	69.58	72.88	70.06	72.50	63.36	61.09	69.69	71.26	**73.41**
		Water	83.22	83.78	79.00	83.21	77.32	65.22	74.86	82.59	**85.00**
		Barren	42.11	38.35	39.67	43.61	47.96	14.61	47.48	47.62	**53.27**
		Forest	57.57	59.34	58.99	**61.52**	56.23	53.89	53.51	54.80	60.14
		Agricultural	71.93	70.60	69.56	69.41	66.97	62.23	68.50	67.90	**75.62**
		Background	71.17	71.27	70.05	70.88	69.66	67.19	69.71	70.05	**72.28**
	MeanF1		67.87	67.94	65.29	68.56	65.00	55.82	65.76	67.80	**71.71**
	mIoU		52.73	53.00	50.33	53.35	48.84	40.39	49.97	52.45	**56.92**
	OA		71.41	71.40	69.34	71.16	68.27	63.25	68.81	69.88	**73.87**

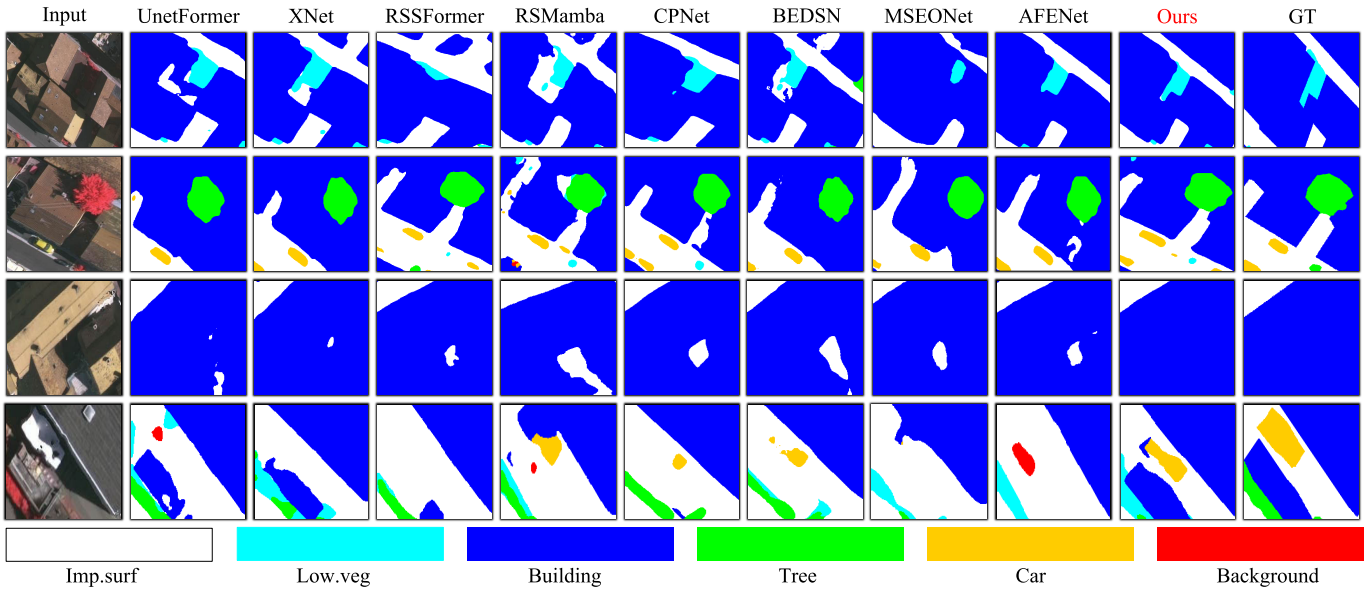


Fig. 3. Local zoomed-in view of the segmentation results on the Vaihingen dataset of PAFCNet and other comparison models. The samples in the figure exhibit abrupt illumination changes induced by building projections.

true object edges both exhibit high-magnitude responses in signal intensity, other models struggle to distinguish their physical attributes-based solely on intensity information, resulting in semantic misclassification. In contrast, by modeling local gradient statistical properties, PAFCNet effectively distinguishes pseudoedge interference caused by illumination projections from true physical structures, thereby achieving correct identification of object categories across various complex lighting scenarios.

2) Results of the LoveDA Dataset: To further verify the generalization ability of PAFCNet, we conducted a quantitative analysis on the more challenging LoveDA dataset. Compared to Vaihingen, which focuses on relatively singular urban

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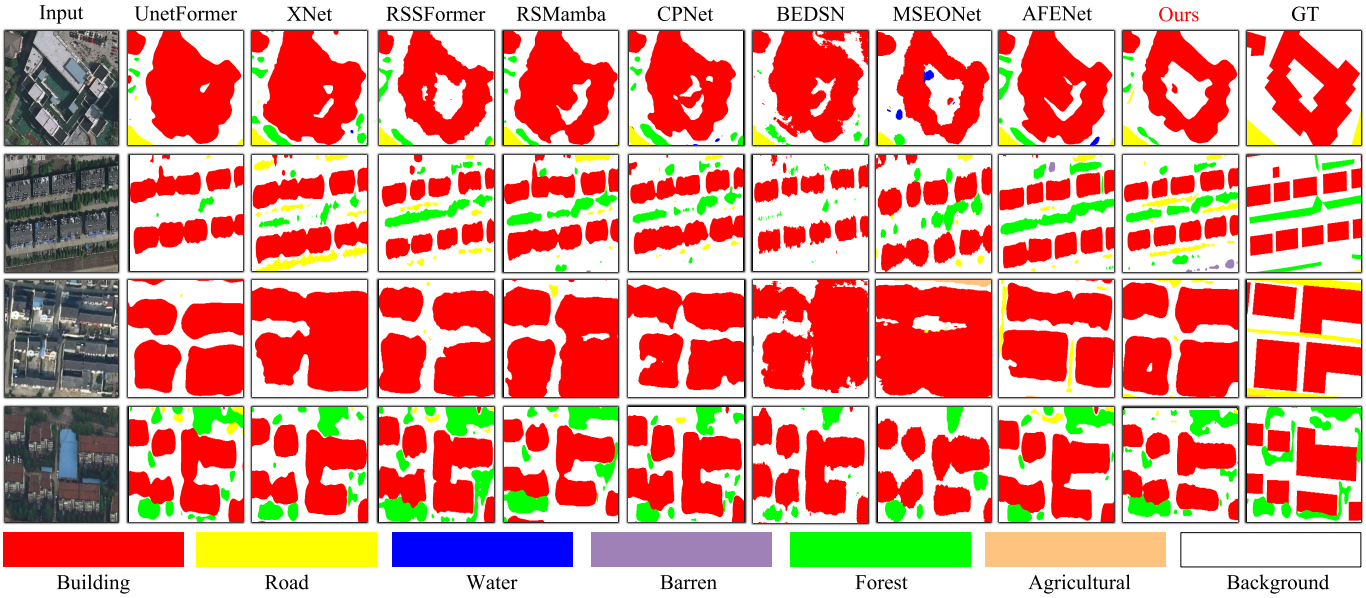


Fig. 4. Local zoomed-in view of the segmentation results on the LoveDA dataset of PAFCNet and other comparison models. The first three rows of the figure display shadows cast by dense building clusters, while the fourth row presents mixed shadows originating from buildings and vegetation.

TABLE II
COMPARISON OF PERFORMANCE ACROSS DIFFERENT MODELS

Method	Params(M)	FLOPs(G)	Time(ms)
UnetFormer [53]	**11.68**	46.96	**11.97**
XNet [11]	326.18	**5343.39**	1269.87
RSSFormer [19]	31.89	336.67	117.44
RSMamba [29]	40.73	149.11	134.97
CPNet [54]	36.59	304.62	60.19
BEDSN [10]	21.25	980.46	179.45
MSEONet [9]	54.13	516.79	164.93
AFENet [36]	20.22	102.35	125.29
PAFCNet(Ours)	28.42	131.94	89.15

environments, LoveDA's cross-regional, dual urban-rural landscape setting encompasses more complex urban landscapes and denser man-made buildings. This brings more severe challenges of intensity variations caused by complex illumination and ground objects. Even under such complex conditions, PAFCNet's performance remains stable. As shown in Table I, PAFCNet significantly outperforms other networks on mean $F1$, mIoU, and OA, reaching 71.71%, 56.92%, and 73.87%, respectively. As seen in the visualization results (Fig. 4), other methods are unable to correctly distinguish the different physical semantics when processing sharp shadows and object edges, resulting in semantic confusion. This confusion visually manifests (as shown in the first, second, and third rows) as erroneous adhesions in the segmentation results, especially the misclassification of shadow regions as part of a building or object. Furthermore, as shown in the fourth row, even in the more complex scenario of mixed shadows cast by buildings and vegetation, our method effectively maintains accurate semantic recognition. These results indicate that PAFCNet can better distinguish shadow-induced pseudoboundaries from true object edges, thereby ensuring the morphological integrity of objects and semantic accuracy.

3) Results of the Potsdam Dataset: To comprehensively validate that PAFCNet not only effectively handles projection interference but also maintains stability under significant intensity variations induced by the ground objects themselves, we conducted comparative experiments on the Potsdam dataset. As shown in Table I, on the Potsdam test set, PAFCNet's $F1$, mIoU, and OA reached 93.26%, 87.57%, and 91.88%, respectively, with all metrics significantly surpassing the other compared methods. Compared to XNet, which also employs frequency-domain information for segmentation, PAFCNet is higher by 1.21%, 2.06%, and 1.32% in the key metrics of $F1$, mIoU, and OA. This advantage fully demonstrates that PAFCNet effectively resolves the semantic ambiguity induced by significant intensity variations, originating not only from illumination projections but also from true ground object structures.

As illustrated in Fig. 5, PAFCNet also exhibits excellent performance when handling ground objects with varying orientations, shapes, and scales. This advantage is primarily attributed to the FAM module discarding fixed frequency domain analysis patterns. Addressing the rich high-frequency textures and fine geometric structures presented by the Potsdam dataset due to its low sampling distance, the FAM introduces a frequency-adaptive mechanism based on geometric alignment via ATs and PKFs. By dynamically responding according to the frequency characteristics of local textures, it improves the capture of fine object details, thereby maintaining stable performance even in complex scenarios characterized by rich high-frequency textures and fine geometric structures.

C. Ablation and Analysis

To assess the validity of the theoretical motivation and the effectiveness of PAFCNet's individual components,

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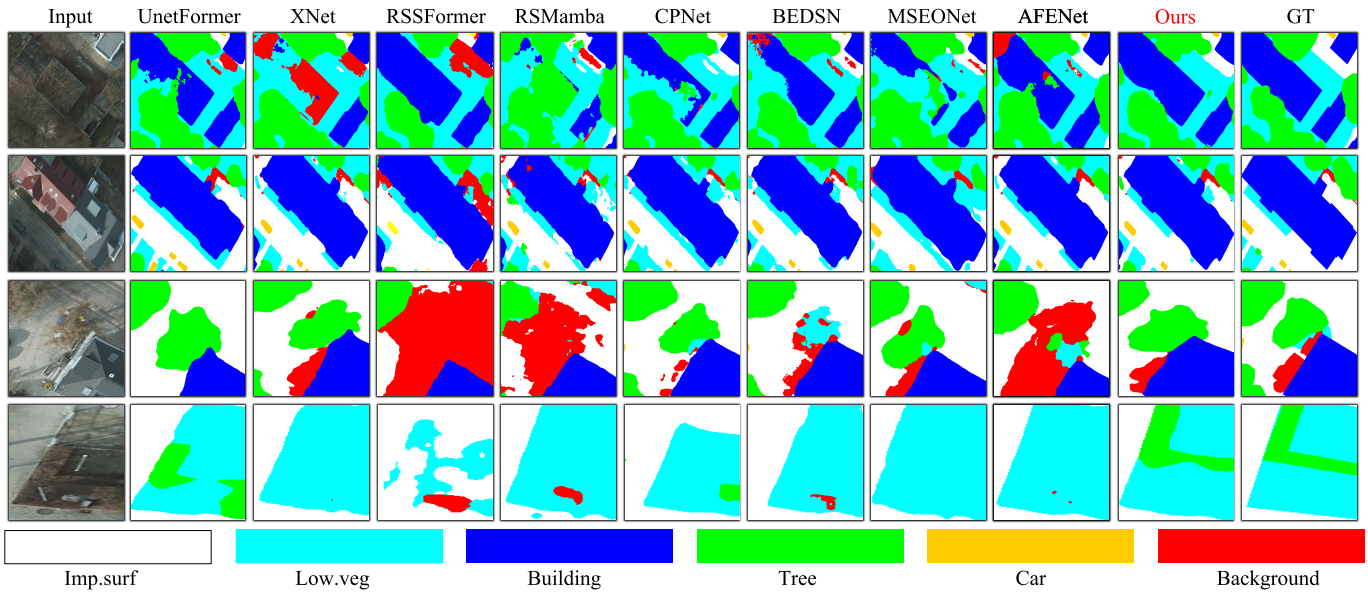


Fig. 5. Local zoomed-in view of the segmentation results on the Potsdam dataset of PAFCNet and other comparison models. The figure highlights regions with significant intensity variations induced by the complexity of ground objects.

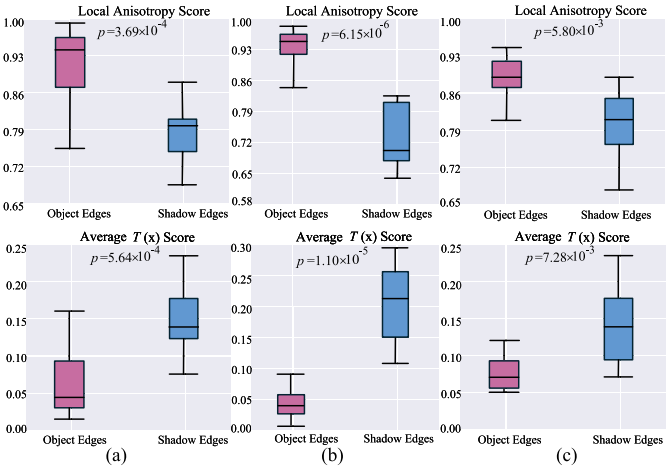


Fig. 6. Statistical validation of gradient anisotropy between object edges and shadow edges. (a) Vaihingen. (b) Potsdam. (c) LoveDA.

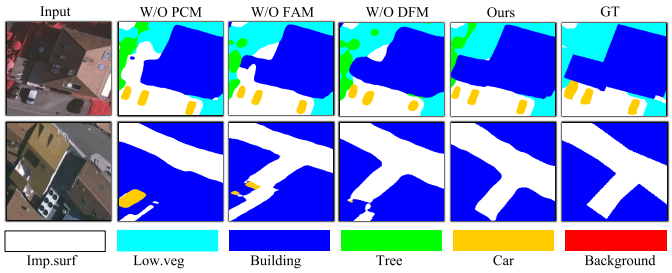


Fig. 7. Visualization of segmentation results after removing individual modules from PAFCNet on the Vaihingen dataset.

we performed experimental evaluations on the Vaihingen, LoveDA, and Potsdam datasets. The experimental results are averaged over multiple trials and analyzed primarily through two performance metrics, mIoU and $F1$.

1) Statistical Validation of Gradient Anisotropy: As shown in Fig. 6, we further evaluate the statistical trend on Vaihingen, Potsdam, and LoveDA, which contain different materials, illumination conditions, shadow directions, and texture patterns. Across the three datasets, real object edges consistently exhibit higher local anisotropy scores, while sharp shadow edges show higher $T(x)$ values. The Mann–Whitney U test confirms that these differences are statistically significant in all datasets ($p < 0.01$). These results support the anisotropy-based motivation of PCM under diverse real remote sensing conditions.

2) Components of PAFCNet: To investigate the role of each component in PAFCNet, we conducted a series of experiments. Table III displays the experimental results after module-wise ablation, where PCM represents the PCM, DFM represents the Dual DFM, and FAM represents the FAM. In the experiments, (W/O) indicates the removal of a component, while (+) indicates its inclusion. Fig. 7 presents the segmentation performance after deleting a single module of PAFCNet, demonstrating that removing any component negatively impacts network performance. To comprehensively evaluate the effectiveness of PCM, FAM, and DFM against representative alternative implementations, we conducted ablation and replacement experiments on the Vaihingen and Potsdam datasets based on a baseline model consisting of two convolutional layers attached directly after the ConvNeXt feature extraction network and a segmentation head, where PCM, FAM, and DFM were individually inserted before the segmentation head, and PCM/FAM were further compared with Sobel gradient response (SOB), Canny edge response (CAN), anisotropy-based modeling (ANI), DWT, DCT, Gabor filtering, AT, and PKFs, as reported in Table IV. Fig. 8 shows the segmentation results after sequentially adding each

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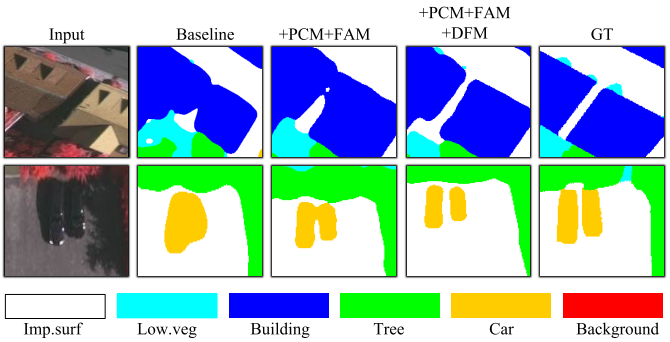


Fig. 8. Visualization of segmentation results after sequentially adding PCM, FAM, and DFM to the baseline.

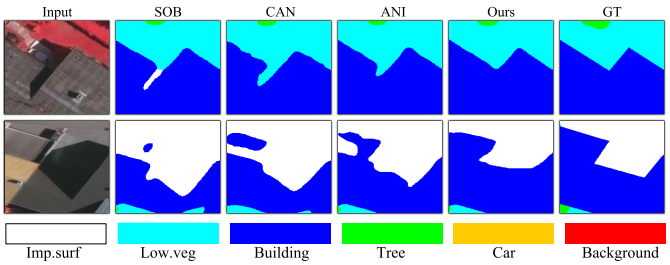


Fig. 9. Visualization comparison of PCM and representative gradient-based alternatives.

TABLE III
RESULTS OF PAFCNET WITH INDIVIDUAL COMPONENTS REMOVED. THE BEST RESULTS ARE INDICATED IN BOLD BLACK

Config	Vaihingen		Potsdam	
	mIoU(%)	F1(%)	mIoU(%)	F1(%)
w/o DFM	83.93	91.16	87.06	92.97
w/o PCM	83.99	91.19	86.84	92.85
w/o FAM	84.05	91.22	86.89	92.87
w/o $\mathcal{L}_{ce}$	84.08	91.24	87.10	92.99
w/o $\mathcal{L}_{dice}$	84.28	91.36	87.32	93.11
PAFCNet	**84.70**	**91.62**	**87.57**	**93.26**

TABLE IV
ABLATION AND REPLACEMENT COMPARISON OF DIFFERENT MODULE VARIANTS IN PAFCNET

Config	Vaihingen		Potsdam	
	mIoU(%)	F1(%)	mIoU(%)	F1(%)
Baseline	82.64	90.38	85.35	91.97
+ DFM	83.72	91.02	86.93	92.89
+ PCM	**83.55**	**90.92**	**86.37**	**92.56**
+ SOB	83.09	90.64	85.91	92.31
+ CAN	82.84	90.48	86.14	92.44
+ ANI	83.18	90.70	86.08	92.41
+ FAM	**83.46**	**90.87**	**86.31**	**92.51**
+ DWT	82.90	90.52	85.96	92.33
+ DCT	82.99	90.58	86.05	92.39
+ GAB	82.95	90.55	85.93	92.32
+ AT	83.30	90.77	86.18	92.46
+ PKF	83.21	90.71	86.24	92.50

component to the baseline model, and comprehensive experimental data indicate that each component we proposed can significantly improve model performance.

3) Impact of PCM: As shown in Table IV, when the PCM is added to the baseline model, the mIoU on the Vaihingen dataset improves by 0.91% and the $F1$ by 0.54%, while on the Potsdam dataset, mIoU and $F1$ increase by 1.02% and 0.59%, respectively. Moreover, compared with representative alternatives based on traditional gradient responses and anisotropy modeling, including SOB, CAN, and ANI, PCM consistently achieves the best performance on both datasets. Specifically, on Vaihingen, PCM outperforms the best alternative ANI by 0.37% in mIoU and 0.22% in $F1$, while on Potsdam, it surpasses the best alternative CAN by 0.23% in mIoU and 0.12% in $F1$. These results indicate that the performance gain of PCM does not arise from general gradient enhancement or ordinary architectural replacement, but from its targeted design for regions with significant intensity variations in remote sensing images.

To further illustrate the qualitative advantage of PCM over traditional alternatives, Fig. 9 presents the corresponding visual comparison. Traditional gradient-based or edge-response-based methods (SOB and CAN), as well as the basic ANI, are prone to producing pseudoedge responses in shadow-covered regions, leading to boundary displacement and misclassification. Specifically, SOB produces obvious slender false detections inside the target, while CAN and ANI also exhibit boundary displacement and class erosion. Moreover, these methods tend to misinterpret shadow edges as physical edges, confusing the model's judgment of real object edges. In contrast, PCM more effectively suppresses pseudoedge responses in shadow regions while preserving real object structural information, yielding better contour continuity and region integrity. These observations suggest that PCM improves discrimination by transforming local gradient spectral differences into confidence estimation for physical structures, rather than simply enhancing local gradient responses or relying on static anisotropy measures.

Beyond the above comparison with traditional alternatives, Fig. 10 further shows the independent contribution of PCM. Although FAM alone improves the sensitivity to regions with significant intensity variations, it still fails to effectively distinguish sharp shadows from real object edges. After introducing PCM, the shadow-induced object adhesion is clearly alleviated. This suggests that merely enhancing the response to regions with significant intensity variations is still insufficient for distinguishing sharp shadows from real object edges, whereas PCM provides more discriminative structural cues for this task. To further reveal the effect of PCM, as illustrated in Fig. 11, we visualize its intermediate features. Fig. 11(b) shows the initial feature x_1, where sharp shadows and object edges exhibit similar intensity responses. Fig. 11(c) shows the rectified feature after PCM, where shadow responses are suppressed while real object structures are largely preserved, indicating that PCM can reduce shadow-induced pseudoedge interference through structural confidence estimation. Fig. 11(e) presents the final fused features, in which the edge responses become more continuous and refined, benefiting

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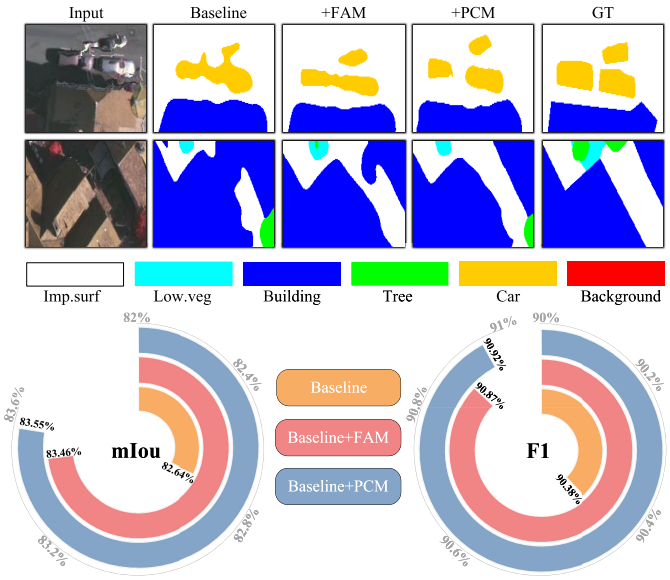


Fig. 10. Visualization of segmentation results after adding PCM and FAM to the baseline.

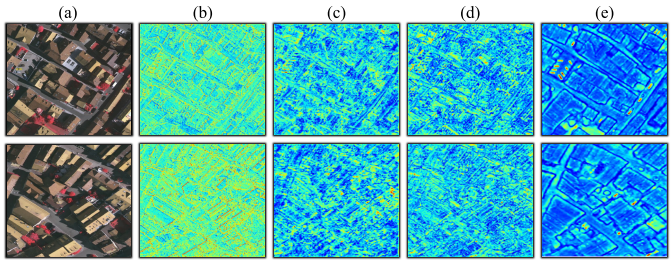


Fig. 11. Feature maps before and after using PCM, FAM, and DFM. (a) Remote sensing image. (b) Feature map x_1. (c) Feature map after applying PCM. (d) Feature map after the element-wise multiplication of FAM and PCM. (e) Feature map processed by PCM, FAM, and DFM.

TABLE V
SENSITIVITY ANALYSIS OF PCM SETTINGS

Setting	Value	mIoU(%)	F1(%)
Default	5×5, $\epsilon = 10^{-6}$, $\tau_0 = 0.03$	**83.55**	**90.92**
Window	3×3	83.47	90.88
Window	7×7	83.52	90.91
ϵ	10^{-8}	83.54	**90.92**
ϵ	10^{-4}	83.53	90.89
τ_0	0.01	**83.55**	**90.92**
Gaussian noise std	0.05	83.52	90.90

from the joint effect of PCM-based shadow suppression and FAM-based frequency detail enhancement.

To further examine the robustness of PCM, we conduct a sensitivity analysis on its key settings, including the local statistical window size, the numerical stability coefficient ϵ, the initial learnable threshold τ_0, and the input noise level. During each test, only one factor is changed while the other training settings are kept unchanged. The default configuration uses a 5×5 window, $\epsilon = 10^{-6}$, and $\tau_0 = 0.03$. The results in Table V show that the performance varies only slightly under different settings, indicating that PCM is not sensitive to these hyperparameters and can provide stable performance.

4) Impact of FAM: As shown by the data in Table III, removing the FAM from the PAFCNet model results in a 0.65% decrease in mIoU and a 0.40% decrease in the $F1$ on the Vaihingen dataset, while on the Potsdam dataset, the corresponding mIoU and $F1$ fall by 0.68% and 0.39%, respectively. Conversely, when the FAM is added to the baseline model, the mIoU on the Vaihingen dataset improves by 0.82% and the $F1$ by 0.49%. On the Potsdam dataset, the mIoU and $F1$ see improvements of 0.96% and 0.54%, respectively. To further evaluate whether this improvement comes from the specific design of FAM rather than from general frequency enhancement alone, Table IV compares FAM with several classical frequency alternatives, including DWT, DCT, and Gabor filtering. The complete FAM consistently achieves the best performance on both datasets. In particular, on Vaihingen, FAM improves over the best classical alternative, DCT, by 0.47% in mIoU and 0.29% in $F1$, while on Potsdam the corresponding gains are 0.26% and 0.12%. These results indicate that the advantage of FAM does not come merely from introducing frequency processing, but from its stronger ability to adapt to the multidirectional frequency patterns and rich texture details of remote sensing images.

We further perform a disentangled ablation on the two key components of FAM, namely, AT and PKF. The results show that both AT and PKF individually retain part of the performance gain, but neither of them alone matches the complete FAM. Specifically, on Vaihingen, the best partial variant AT still remains 0.16% lower than the complete FAM in mIoU and 0.10% lower in $F1$, while on Potsdam the best partial variant PKF remains 0.07% lower in mIoU and 0.01% lower in $F1$. This suggests that the effectiveness of FAM is not solely due to affine-aligned sampling or parameterized sinusoidal filtering alone, but arises from the complementary effect of geometry-adaptive alignment and learnable frequency modeling.

Fig. 10 visually reveals the effectiveness of the FAM. After applying the FAM, the overall contours of the objects are restored and sharpened, benefiting from its ability to better capture fine-grained textural and geometric details. However, the two vehicles at the bottom of the first row remain adhered, since both sharp shadows and true edges may produce similar strong responses, making it difficult for the FAM alone to distinguish them effectively. In contrast, after applying the PCM, the shadow-induced adhesion between vehicles is largely alleviated, achieving successful individual separation. This demonstrates that statistical decisions based on physical genesis can effectively suppress shadow interference. At the same time, it is also observed that the object contours generated using only the PCM are slightly inferior to those of the FAM in terms of refinement. This comparison indicates that the FAM is more effective in modeling fine-grained structural details, while the PCM specializes in suppressing physical interference, making the combination of the two crucial for segmentation in complex scenes.

5) Impact of DFM: As shown by the data in Table III, removing the DFM from the PAFCNet model results in a 0.77% decrease in mIoU and a 0.46% decrease in the $F1$ on the Vaihingen dataset, while on the Potsdam dataset, the

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TABLE VI
SHADOW-AWARE EVALUATION ON THE CONSTRUCTED SHADOW SUBSET

Method	$\mathrm{mIoU}_{sh}\uparrow$	$\mathrm{mIoU}_{non}\uparrow$	$\mathrm{Err}_{sh\text{-}bd}\downarrow$	FAD$\downarrow$
Baseline	53.40	67.66	23.74	25.64
Baseline+PCM	56.35	69.26	22.97	25.33
AFENet	57.11	70.16	22.55	24.73
PAFCNet	**59.56**	**71.46**	**21.87**	**24.08**

corresponding mIoU and $F1$ fall by 0.51% and 0.29%, respectively. Conversely, when the DFM is added to the baseline model, the mIoU on the Vaihingen dataset improves by 1.08% and the $F1$ by 0.64%. On the Potsdam dataset, the mIoU and $F1$ see improvements of 1.58% and 0.92%, respectively.

Fig. 8 visually validates the effectiveness of the DFM. In the output of the model equipped only with the FAM and PCM, some local adhesion regions persist at the object edges, which indicates that features from different sources have not been optimally fused. After the introduction of the DFM, these adhesion regions are precisely alleviated. This is attributed to the deep interaction of heterogeneous features within a single pixel facilitated by the DFM, which better exploits their complementarity, thereby alleviating semantic ambiguity.

6) Shadow-Aware Evaluation: To further verify whether PAFCNet reduces shadow-induced segmentation errors, we evaluate different variants on the constructed shadow-aware subset. As shown in Table VI, PAFCNet achieves the best performance on both shadow and nonshadow regions. Compared with the baseline, PAFCNet improves mIoU_{sh} from 53.40% to 59.56% and reduces $\mathrm{Err}_{sh\text{-}bd}$ from 23.74% to 21.87%. Meanwhile, FAD decreases from 25.64% to 24.08%, indicating that PAFCNet can better suppress false attachment of shadow regions to adjacent objects.

7) Impact of the Loss Function: As can be seen from the experimental results in Table III, different configurations of the loss function have varying degrees of impact on model performance. On the Vaihingen dataset, removing $\mathcal{L}_{ce}$ causes mIoU to decrease by 0.62% and $F1$ by 0.38%, while removing $\mathcal{L}_{dice}$ loss leads to a 0.42% decrease in mIoU and a 0.26% decrease in $F1$. On the Potsdam dataset, removing $\mathcal{L}_{ce}$ results in a 0.47% drop in mIoU and a 0.27% drop in $F1$, whereas removing $\mathcal{L}_{dice}$ has a smaller effect, with mIoU decreasing by 0.25% and $F1$ by 0.15%. Removing $\mathcal{L}_{ce}$, which is used to optimize pixel-level classification and boundary distinction, significantly reduces the mIoU and $F1$-scores. In contrast, removing $\mathcal{L}_{dice}$, which primarily addresses the issues of region overlap and class imbalance, has a lesser impact, indicating that its role is more auxiliary. Overall, combining $\mathcal{L}_{ce}$ and $\mathcal{L}_{dice}$ balances the needs of interclass discrimination and regional overlap optimization. Removing any of these loss functions will negatively affect model performance to some extent.

8) Limitation Analysis on Low-Contrast Edges: To clarify the applicable scope of PCM, we further analyze its behavior under low-contrast edge conditions. As shown in Fig. 12, when only PCM is introduced, the predictions still suffer from incomplete edge continuity in low-contrast regions. The full model can partially alleviate this issue by incorporating complementary features from other modules. In more extreme cases, such as textureless or weakly textured surfaces, or highly aligned background textures, shadow edges may form local gradient structures similar to real edges, causing biased discrimination when relying only on PCM. These observations indicate that the effectiveness of PCM depends on clear structural gradients and distinguishable anisotropic cues introduced by local texture perturbations. When these cues are weakened, the discriminative ability of PCM becomes limited.

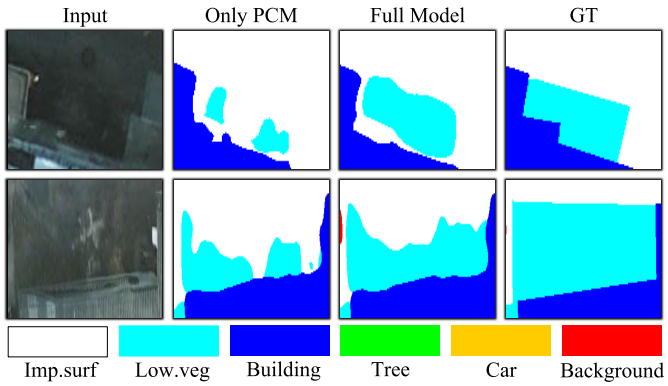


Fig. 12. Visualization of a low-contrast edge case.

V. CONCLUSION

In this article, we propose PAFCNet to address significant intensity variations by explicitly modeling their distinct physical causes. Rather than relying solely on intensity variations, we analyze the differences in local gradient statistics between real object edges and sharp shadow edges. Based on this observation, PCM formulates anisotropy analysis as a differentiable statistical decision process, improving the robustness of physical priors in real-world scenes. On this basis, FAM further models textural and geometric details to enhance the model's perception of regions with significant intensity variations, while DFM facilitates effective interaction between heterogeneous features to reduce semantic ambiguity. Extensive experiments on multiple benchmark datasets show that PAFCNet consistently outperforms state-of-the-art methods, demonstrating the value of incorporating physical priors into remote sensing image segmentation. We hope this study offers a useful perspective for future research on integrating physical priors with deep learning.

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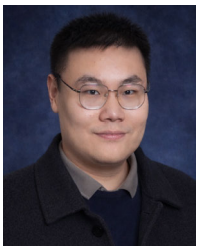
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